

Lee Foster

1.6) Q.

The heat flux through a wooden slab 50mm thick whose inner & outer surface temperatures are 40°C and 20°C respectively, has been determined to be 40 W/m^2 .

What is the thermal conductivity of the wood?

Given:

$$\text{heat flux: } q_x'' = k \frac{\partial T}{\partial x} = 40 \text{ W/m}^2$$

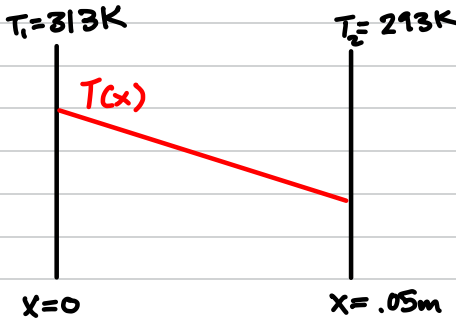
$$L_x = 50 \text{ mm} = .05 \text{ m}$$

$$T_1 = 40^\circ\text{C} = 313 \text{ K}, \quad T_2 = 293 \text{ K}$$

inner surface outer surface

find: K_1

Illustration



assumptions:

Steady state
no heat gen or loss
constant thermal cond.
Linear temp. Profile

Solution:

$$q_x'' = 40 \text{ W/m}^2 = -K_1 \frac{\Delta T}{L} = -K_1 \frac{293 \text{ K} - 313 \text{ K}}{.05 \text{ m}} = K_1 \frac{20 \text{ K}}{.05 \text{ m}}$$

$$40 \frac{\text{W}}{\text{m}^2} = K_1 \frac{20 \text{ K}}{.05 \text{ m}} \Rightarrow K_1 = .1 \frac{\text{W}}{\text{mK}}$$

1.14) A nonuniform wall with thermal conductivity follows $K = ax + b$, where a & b are constants. the heat flux is known to be constant.

Determine expressions for the temperature gradient and the temperature Distribution when the temperature at $x=0$; $T=T_1$.

given:

$$K_1 = ax + b ; a \text{ \& } b \text{ are constant}$$

$$q_x'' = \text{constant}$$

$$T(x=0) = T_1$$

assumptions:

Steady state

1 Dimensional

find: Temp Gradient, $\frac{dT}{dx}$. & $T(x)$

Solution:

$$q_x'' = -K \frac{dT}{dx} \Rightarrow \frac{dT}{dx} = \frac{q_x''}{-K} = \frac{q_x''}{-(ax+b)}$$

$$\frac{dT}{dx} = - \frac{q_x''}{ax+b}$$

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$$dT = - \frac{q_x''}{ax+b} dx$$

$$T(x) - T_1 = \int_0^x \frac{1}{ax+b} dx (-q_x'') \quad u = ax+b \quad dx = \frac{du}{a}$$

$$\frac{du}{dx} = a$$

$$T(x) - T_1 = - \frac{q_x''}{a} \ln\left(\frac{ax+b}{b}\right)$$

$$T(x) = T_1 - \frac{q_x''}{a} \ln\left(\frac{ax+b}{b}\right)$$

1.15) Q.

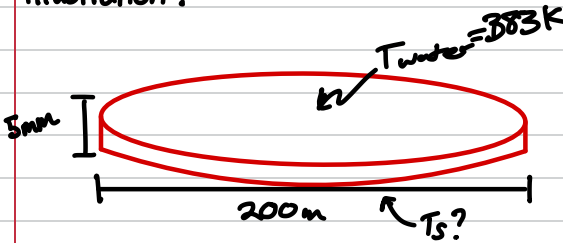
The 5mm thick Bottom of a 200 mm diameter Pan may be made from aluminum ($K=240 \text{ W/mK}$) or copper ($K=390 \text{ W/mK}$). When used to boil water, The bottom surface is exposed to 110°C . If heat is transferred from the stove to the pan at a rate of 600W, what is the temperature of the surface in contact w/ the stove for each material?

Given:

Pan thickness: $5\text{mm} = .005\text{m}$
 Pan diameter: $200\text{mm} = .2\text{m}$
 2 material choices (K changes),
 Copper $\sim K_1 = 390 \text{ W/mK}$
 Aluminum $\sim K_2 = 240 \text{ W/mK}$
 $T_{\text{water}} = 110^\circ\text{C} = 383\text{K}$
 $Q = 600\text{W}$

find: T_{SA} & T_{SC}

illustration:



Calculations:

$$A = \pi r^2$$

$$A = \pi (.1\text{m})^2$$

$$= .0314\text{m}^2$$

$$q''_x = \frac{Q}{A} = \frac{600\text{W}}{.0314\text{m}^2}$$

$$= 19,108 \text{ W/m}^2$$

Copper

$$q''_x = \frac{Q}{A} = \frac{K(T_{SC} - T_{\text{water}})}{L}$$

$$T_{SC} = T_{\text{water}} + \frac{QL}{KA}$$

$$= 383\text{K} + \frac{(600\text{W})(.005\text{m})}{(390 \text{ W/mK})(.0314\text{m}^2)} = 383.24\text{K}$$

$$110.24^\circ\text{C}$$

Aluminum

$$T_{SA} = 383 + \frac{600 \cdot .005}{240 \cdot .0314} = 383.4\text{K}$$

$$110.4^\circ\text{C}$$

1.19) Q.

Air at 40°C flows over a long 25mm diameter cylinder w/ an embedded electrical heater. In a series of tests measurements were made of the power per unit length, P' , required to maintain the cylinder surface temp. at 300°C for different free stream velocities V of the air, w/ these results.

Air velocity, V (m/s)	1	2	4	8	12
Power, P' (W/m)	490	658	983	1507	1963

- a.) Determine the convection coefficient for each velocity & Display results graphically.
 b.) Assuming the dependence of the convection coeff. on the velocity to be of the form $h = CV^n$ determine the parameters C & n from part (a).

a.) given: $T_{\text{air}} = 40^\circ\text{C} = 313\text{K}$
 $D = 25\text{mm} = 0.025\text{m}$
 $T_{\text{sur}} = 300^\circ\text{C} = 573\text{K}$
 find: $h_1, h_2, h_4, h_8, h_{12}$

Calculations:

Wetted Area: $A' = \pi D$
 $= 0.0785\text{m}^2/\text{m}$

$\frac{P'}{A'} = h(\Delta T)$ $\Delta T = 573\text{K} - 313\text{K} = 260\text{K}$

$h_1 = \frac{P'_1}{A' \Delta T} \therefore h \propto P' (A' \cdot \Delta T) = 20.23\text{mK}$

$h_1 = 22.19\text{W/m}^2$ $h_5 = 96.79$

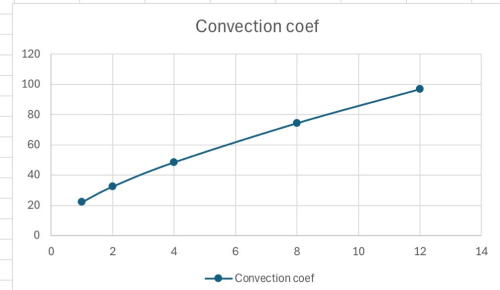
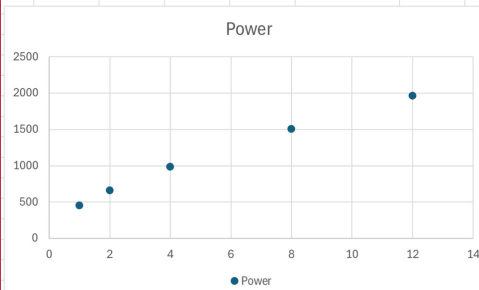
$h_2 = 32.45\text{W/m}^2$

$h_3 = 48.47\text{W/m}^2$

$h_4 = 74.31\text{W/m}^2$

a. continue

Air	Power	Air	Convection coef
1	450	1	22.19
2	658	2	32.45
4	983	4	48.47
8	1507	8	74.31
12	1963	12	96.79



b.) using the power trendline on excel,
 $h = C v^n$ where $C = 22.07$ & $n = .589$